

SIR: A Sparse-Interaction Keystream Generator with a Hardware-Oriented Architecture

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Abstract

Lightweight keystream generators are widely used in resource-constrained digital systems, where implementation efficiency in area, power, and logic structure is a primary design concern. Conventional designs predominantly employ shift-register-based state propagation, in which diffusion is inherently coupled with sequential data movement. This work investigates an alternative architectural approach in which state mixing is achieved through sparse interaction among state variables, enabling a decoupling between diffusion and register propagation. We present *SIR*, a sparse-interaction keystream generator with a 128-bit internal state composed of a nonlinear 64-bit primary state and a 64-bit auxiliary linear state. The primary state is updated using a compact four-input Boolean function applied over a fixed sparse neighbourhood, while the auxiliary state provides lightweight round-dependent perturbation. This structure realizes diffusion through parallel combinational interaction, leading to a distinct hardware profile characterized by reduced reliance on sequential storage and increased distributed logic. The architectural behaviour is evaluated through diffusion and statistical experiments, showing rapid propagation of local perturbations across the state within 14–15 rounds and no observable low-order dependence between internal state variables and output in the tested regime. Hardware implementation on a Xilinx Artix-7 FPGA requires 183 LUTs and 177 flip-flops, while ASIC synthesis using a 45 nm standard-cell library results in an area of 3079 gate equivalents. Comparative evaluation with Grain-128, Trivium, and Espresso under identical implementation conditions demonstrates that the proposed architecture provides a competitive trade-off between combinational logic and sequential resources. The results indicate that sparse-interaction-based state evolution constitutes a viable architectural alternative for lightweight keystream generation, particularly in hardware-oriented and FPGA-based design settings.

Keywords:

Lightweight cryptography, Keystream generator, Sparse state interaction, FPGA-based implementation, Hardware-oriented design, Nonlinear diffusion, Compact cryptographic architectures

1. Introduction

Lightweight keystream generators are widely used in resource-constrained digital systems, where implementation efficiency in area, power, and latency is a primary design concern. Such requirements arise in embedded platforms, Internet-of-Things (IoT) devices, and wireless sensor networks, where hardware resources are tightly limited [1, 2, 3, 4, 5]. In these settings, the design of keystream generators is inherently hardware-oriented, requiring an effective balance between state-update complexity, diffusion behaviour, and efficient realization in digital logic.

A large class of existing lightweight stream cipher designs follows a shift-register-oriented paradigm. Representative constructions, including Grain-128, Trivium, and Espresso, employ combinations of linear and nonlinear feedback shift registers together with filtering or feedback functions to generate keystream sequences [6, 7, 8, 9, 10]. These designs are well suited to hardware implementation because shift-register structures map naturally to chains of flip-flops with limited combinational logic, resulting in compact, regular, and predictable implementations.

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A defining characteristic of this paradigm is that diffusion is achieved through sequential propagation of state information. Local perturbations are transported along the register over multiple clock cycles, and the rate at which information spreads is therefore bounded by the underlying shift structure [11, 12, 13]. Consequently, diffusion and storage are structurally coupled: the same register elements that store the state also realise its mixing, leading to a tightly constrained relationship between state-update behaviour and hardware organisation.

This observation raises a fundamental architectural question for hardware-oriented design: whether effective mixing can be achieved without relying on sequential propagation. In particular, it is of interest to explore keystream-generation mechanisms in which diffusion is realised through local interaction among state variables within a single clock cycle, thereby enabling alternative trade-offs between combinational logic and sequential storage.

In this work, we investigate an alternative approach based on sparse interaction among state elements. Instead of propagating information through register shifting, the proposed architecture employs a compact nonlinear update rule in which each state element interacts with a small neighbourhood of neighbouring positions. When applied uniformly across the state, this local interaction structure enables parallel combinational mixing, allowing perturbations to spread through repeated iteration of simple local rules.

From a hardware perspective, this approach leads to a distinct architectural organisation. Diffusion is carried out through combinational interaction, while flip-flops primarily serve as storage elements between rounds. As a result, the implementation exhibits a different distribution of resources compared to conventional shift-register-based designs, with reduced reliance on sequential state and increased use of distributed logic. This characteristic is particularly relevant for FPGA implementations, where logic and storage resources incur different costs and routing implications.

Motivated by this perspective, we introduce *SIR* (Sparse Interaction Register), a lightweight keystream generator with a hardware-oriented architecture. The construction consists of a 64-bit nonlinear primary state updated through a compact four-input Boolean function applied over a sparsely connected neighbourhood, coupled with a 64-bit auxiliary LFSR that provides round-dependent perturbation. This combination forms a structured interaction network that supports iterative mixing while maintaining a compact and implementable design.

The behaviour of the proposed architecture is evaluated through both structural analysis and hardware implementation. The analysis examines diffusion and statistical properties using differential propagation, correlation screening, and information-theoretic measurements, providing insight into state evolution under the proposed update mechanism. Complementary hardware evaluation is performed through FPGA implementation and ASIC synthesis, enabling quantitative comparison of area and resource utilisation against established lightweight keystream generators.

The objective of this study is twofold. First, it introduces a keystream generation architecture in which diffusion is achieved through parallel sparse interaction rather than sequential propagation. Second, it characterises the resulting hardware implementation profile through empirical evaluation. Together, these results demonstrate that sparse-interaction-based designs provide a viable and practically relevant alternative within the hardware-oriented keystream generation design space.

The main contributions of this work are summarized as follows:

1. **Hardware-oriented keystream generator architecture.** We introduce *SIR* (Sparse Interaction Register), a lightweight keystream generator based on a 128-bit internal state comprising a nonlinear 64-bit primary register and a 64-bit auxiliary LFSR, coupled through a lightweight injection mechanism suitable for hardware implementation.
2. **Sparse interaction-based state update mechanism.** The proposed architecture employs a compact nonlinear update rule defined over local neighbourhoods of the state, enabling parallel information propagation through sparse interaction rather than sequential shift-register movement, thereby providing an alternative organisation of diffusion in hardware.
3. **Experimental characterization of structural behaviour.** The mixing and statistical properties of the architecture are evaluated using differential propagation analysis, linear-correlation screening, and state-output mutual-information measurements, supporting assessment of the proposed update structure.
4. **FPGA and ASIC implementation analysis.** The design is described in Verilog and evaluated on a Xilinx Artix-7 FPGA and through ASIC synthesis using the Nangate 45 nm standard-cell library, reporting resource utilisation and area metrics under a consistent implementation flow.

5. **Comparative hardware evaluation.** The proposed architecture is compared with representative lightweight keystream generators, including Grain-128, Trivium, and Espresso, highlighting a distinct trade-off between sequential storage and combinational mixing in practical implementations.

The remainder of this paper is organized as follows. Section 2 reviews prior work on lightweight stream cipher design and hardware-oriented constructions. Section 3 introduces the proposed SIR. Section 4 presents experimental evaluation of diffusion and statistical behaviour. Section 5 reports FPGA and ASIC implementation results. Section 6 provides additional security analysis. Section 7 discusses broader implications. Finally, Section 8 concludes the paper.

2. Related Work

Lightweight keystream generation has been studied extensively due to its importance in constrained digital platforms such as embedded systems, wireless sensor networks, and Internet-of-Things (IoT) devices [11, 14]. In such environments, cryptographic modules must operate under strict constraints on silicon area, power consumption, memory footprint, and latency, while maintaining acceptable security properties. As a result, a substantial body of work has focused on keystream-generation mechanisms specifically tailored for efficient hardware realisation.

2.1. The Shift-Register Paradigm and Its Hardware Characteristics

The dominant architectural class in lightweight keystream generation is based on linear and nonlinear feedback shift-register (LFSR/NLFSR) structures [6, 7, 8, 15, 16]. Representative designs include Grain-128, which combines an LFSR and an NLFSR with a nonlinear output function [6], and Trivium, which employs three interacting shift registers with nonlinear feedback [7]. More recent designs such as Espresso explore optimised nonlinear feedback structures while retaining the shift-register framework as the primary mechanism for state evolution [8].

The hardware attractiveness of these designs follows directly from their structural organisation. Shift-register-based state updates map naturally to chains of flip-flops with a small number of tap-XOR or nonlinear feedback operations per cycle. This produces a characteristic hardware profile: relatively deep sequential state (large flip-flop count), shallow combinational paths per round, and predictable timing closure behaviour [14, 17].

A critical structural property of this paradigm is that diffusion is achieved through sequential propagation. Local state changes are transported across the register over multiple clock cycles, and the elements responsible for this propagation are precisely the flip-flops that dominate the hardware area budget. As a result, diffusion and storage are intrinsically coupled: improving diffusion requires more propagation, and propagation requires sequential state.

A substantial literature has examined the cryptanalytic and implementation properties of these constructions [18, 19, 20, 21]. Algebraic structure, differential characteristics, and time–memory–data trade-offs have all been analysed in detail [22, 23, 24]. These studies reinforce a consistent observation: the efficiency of shift-register-based designs is achieved through a tight coupling between their diffusion mechanism and their sequential hardware structure.

2.2. Hardware Optimisation Within the Shift-Register Framework

Within this paradigm, significant effort has been devoted to reducing implementation cost while preserving security properties. FPGA and ASIC comparative evaluations have been reported for a wide range of designs in order to assess their practical efficiency under common synthesis conditions [14, 25, 17]. These studies typically evaluate logic utilisation, flip-flop count, maximum operating frequency, and power consumption, and establish a well-characterised body of reference implementation data for the shift-register class.

The relationship between internal update structures and hardware cost has been examined in greater detail [2, 11]. In particular, the balance between combinational logic and sequential storage has been identified as a key factor influencing area, routing complexity, and timing closure. Designs that minimise combinational logic tend to rely heavily on registers, while those that increase nonlinear mixing per round incur higher combinational cost [8, 2].

Despite these variations, such optimisations remain constrained by the same underlying principle: diffusion is governed by sequential propagation, and the storage elements that enable this propagation continue to dominate the hardware budget. Consequently, optimisation within this framework explores different points along a fixed design axis rather than altering the fundamental relationship between diffusion and storage.

2.3. Alternative State-Update Mechanisms: Cellular Automata and Locally Connected Systems

Motivated by the limitations of sequential propagation, alternative update mechanisms based on local interaction have been explored. Cellular automaton (CA)-based keystream generators provide a prominent example, in which each state element is updated as a Boolean function of a small neighbourhood, with all updates applied in parallel in each round [26, 27, 28].

Constructions such as CAR30 [26] and CAStream [27] demonstrate that rapid diffusion can emerge from repeated application of local nonlinear rules without relying on long-range shift-register propagation. This establishes that parallel local interaction is a viable mechanism for achieving state mixing.

However, purely autonomous CA-based generators have been shown to exhibit exploitable correlation properties under certain conditions [29]. This limitation highlights the difficulty of achieving both strong diffusion and sufficient temporal variation within a fully autonomous local-update system.

The construction proposed in this work shares the use of uniform local Boolean updates applied in parallel across the state. However, it differs structurally by introducing an independent auxiliary LFSR that injects round-dependent perturbations into the nonlinear state. This coupling prevents autonomous evolution and introduces controlled temporal variation, while also defining a clear hardware partition between a compact sequential backbone and a combinational interaction network.

2.4. The Unexamined Hardware Profile of Sparse-Interaction Architectures

While prior work has established both the efficiency of shift-register-based designs and the diffusion capability of locally connected update rules, the hardware implications of replacing sequential propagation with parallel sparse interaction have not been systematically characterised.

In shift-register-based architectures, the same elements that store the state also realise diffusion through propagation along the register chain. In contrast, in a sparse-interaction architecture, diffusion is achieved combinatorially through parallel application of a local Boolean rule, while flip-flops serve only as storage between rounds. This introduces a structural separation between the diffusion mechanism and the storage mechanism.

The consequences of this separation are not directly predictable from existing literature. In particular, it is unclear how the redistribution of cost between combinational logic and sequential storage affects overall implementation efficiency under realistic FPGA and ASIC synthesis conditions, and whether this redistribution yields a useful design point for resource-constrained environments.

The present work addresses this gap through the design and systematic evaluation of SIR, a keystream generator in which sparse parallel interaction replaces shift-register propagation as the primary diffusion mechanism. The empirical analysis of Section 4 and the hardware evaluation of Section 5 together provide a first characterisation of this architectural class from both structural and implementation perspectives.

2.5. Selection of Reference Designs and Scope of Comparison

For hardware evaluation and structural comparison, three representative lightweight keystream generators were selected: Grain-128 [6], Trivium [7], and Espresso [8].

The first criterion is architectural representativeness. These designs span the range of hardware profiles within the shift-register paradigm, from register-dominated to more combinatorially intensive configurations. Grain-128 and Trivium are widely used reference points in hardware evaluation studies, while Espresso represents a more recent optimisation within the same architectural framework.

The second criterion is implementation consistency. All designs, including SIR, were implemented in Verilog and synthesised under identical tool flows as described in Section 5, ensuring that the reported results reflect architectural differences rather than tool-dependent artefacts.

The comparison is not intended to establish uniform superiority of the proposed design. Instead, it is structured to characterise how a sparse-interaction architecture positions itself within the LUT-FF design space relative to the shift-register paradigm. Because diffusion is realised differently, SIR is expected to exhibit a distinct trade-off profile, with reduced sequential state and increased combinational mixing.

2.6. Positioning of the Present Work

The present work introduces and evaluates a keystream generator in which diffusion is achieved through parallel sparse interaction rather than sequential propagation. This distinguishes it from both shift-register-based designs and purely autonomous CA-based generators.

The SIR construction employs a 64-bit nonlinear primary state updated through a uniform local Boolean rule, coupled with a 64-bit auxiliary LFSR that provides round-dependent perturbation. This combination enables interaction-driven diffusion while maintaining a controlled and compact sequential structure.

The contribution is therefore architectural and empirical. It demonstrates that sparse-interaction-based update mechanisms can achieve effective diffusion, pass standard statistical screening, and occupy a coherent and potentially useful position in the hardware design space for lightweight keystream generation.

Rather than refining existing designs, the work expands the design space by introducing a configuration in which diffusion and storage are structurally decoupled. The results of Section 5 provide the first systematic evidence of how this separation manifests in practical hardware implementations.

3. SIR Construction

This section presents a complete specification of the proposed keystream generator, *SIR*. The design maintains a structured internal state consisting of a nonlinear primary component and an auxiliary linear component that provides round-dependent perturbations. The construction combines a sparse state-interaction pattern with a lightweight nonlinear Boolean primitive in order to promote mixing while maintaining low hardware complexity.

All arithmetic is performed over $\mathbb{F}_2$. The symbol $\oplus$ denotes bitwise XOR, and juxtaposition denotes logical AND. Unless otherwise stated, indices are taken modulo 64.

The construction is defined through the following components: (i) state representation, (ii) nonlinear mixing primitive, (iii) neighbourhood interaction, (iv) primary state evolution, (v) auxiliary-state-driven perturbation, and (vi) keystream output generation.

3.1. State Representation

SIR maintains a 128-bit internal state partitioned into two 64-bit components:

$$P^{(t)} = (P_0^{(t)}, P_1^{(t)}, \dots, P_{63}^{(t)}), \quad (1)$$

$$L^{(t)} = (L_0^{(t)}, L_1^{(t)}, \dots, L_{63}^{(t)}), \quad (2)$$

where t denotes the round index.

The vector $P^{(t)}$ forms the *primary nonlinear state*. It evolves through a nonlinear transformation and serves as the direct source of the keystream output.

The vector $L^{(t)}$ forms an *auxiliary linear state*. It evolves according to a linear feedback shift register and injects round-dependent perturbations into the primary state. This separation introduces additional temporal variation into the state update and avoids a purely autonomous evolution of the nonlinear component.

3.2. Design Philosophy

The design of SIR follows three guiding principles aimed at balancing mixing behaviour, structural simplicity, and hardware efficiency.

Sparse structured interaction. Instead of relying on dense connectivity among state variables, SIR employs a sparse interaction topology derived from a fixed set of offsets. Each state position interacts with only a small number of neighbours, forming a bounded-degree interaction network. Through repeated rounds, such local interactions can distribute perturbations across the state while keeping hardware fan-in small.

Local nonlinear computation. Nonlinearity is introduced through a compact four-input Boolean primitive (RULEA) applied uniformly across all state positions. Using identical nonlinear units simplifies the hardware architecture and avoids unnecessary structural irregularity. The limited input arity ensures that the nonlinear function can be implemented with a small number of logic gates while still achieving algebraic degree three.

Auxiliary perturbation channel. The auxiliary state component acts as a lightweight perturbation source that injects round-dependent information into the primary nonlinear state. This design introduces additional temporal variation into the update dynamics while keeping the perturbation mechanism simple in hardware. The auxiliary state is implemented as an LFSR so that this mechanism incurs minimal implementation overhead.

Together, these principles define an architecture in which mixing is produced through repeated sparse interactions rather than through large nonlinear components. This design philosophy aligns well with hardware-oriented implementation goals, where logic complexity and routing overhead are often as important as state size.

3.3. Design Objectives and Evaluation Scope

SIR is intended for keystream generation in resource-constrained environments such as embedded systems and FPGA accelerators. The design operates with a 128-bit internal state and a 128-bit secret key, with emphasis on compact hardware realization together with empirically satisfactory mixing and output behaviour.

The threat model assumes an adversary with full knowledge of the generator specification who observes arbitrarily long keystream sequences produced under a fixed secret key and initialization vector. The adversary may attempt to recover internal state information or to distinguish the generated keystream from a truly random sequence using standard cryptanalytic techniques.

Within this scope, the design is intended to reduce the likelihood of the following classes of structural weakness:

- **Poor differential diffusion.** The nonlinear update structure and sparse state-interaction pattern are intended to encourage rapid expansion of state differences across multiple rounds.
- **Strong low-order linear relations.** The nonlinear primitive and the distributed output function are intended to limit simple linear dependencies between internal state bits and the keystream output.
- **Trivial state decoupling.** The auxiliary perturbation mechanism is intended to prevent the nonlinear state evolution from operating as a completely isolated deterministic subsystem.
- **Observable statistical bias.** The output function combines nonlinear sampling with a global parity component in order to reduce obvious statistical imbalance in the generated keystream.

The effectiveness of these design choices is evaluated in the subsequent sections through empirical diffusion experiments, linear correlation analysis, information-theoretic leakage measurements, and comparative hardware evaluation against established keystream generators. Taken together, these results provide a structural evaluation of the construction with respect to mixing behaviour, statistical properties, and observable state–output dependencies.

3.4. Local Nonlinear Primitive

Nonlinear mixing is performed using a single four-input Boolean function denoted RULEA. The function is defined as

$$\text{RULEA}(x_0, x_1, x_2, x_3) = 1 \oplus x_2 \oplus x_0x_2 \oplus x_1x_2 \oplus x_1x_3 \oplus x_0x_2x_3. \quad (3)$$

The algebraic normal form (ANF) contains quadratic and cubic terms, yielding algebraic degree 3. The Walsh spectrum of the function exhibits no dominant coefficient at zero frequency, indicating that the function is unbiased.

The four-input arity was chosen deliberately as a hardware-oriented design constraint. Limiting the fan-in allows each nonlinear update unit to map directly to a small cluster of primitive logic gates, which reduces FPGA routing complexity and ASIC gate count while retaining sufficient nonlinearity for iterative mixing.

3.5. Local Neighbourhood

A key feature of the design is the structured selection of inputs to RULEA. Rather than arbitrary wiring, the neighbourhood of each state position is determined by a fixed generator set

$$\mathcal{S} = \{-1, +1, +8\} \pmod{64}. \quad (4)$$

For position i , the four inputs to RULEA are therefore

$$P_i, P_{i-1}, P_{i+1}, P_{i+8}.$$

The offsets ± 1 introduce immediate local coupling between adjacent positions, while the offset $+8$ creates a longer-range interaction across the state. Together, these edges form a sparse interaction network with bounded degree, allowing perturbations to spread across the state through repeated updates.

Several candidate generator sets were evaluated experimentally. In particular, $\{-1, +1, +2\}$ and $\{-1, +1, +16\}$ produced noticeably slower diffusion and exhibited higher extinction rates when subjected to single-bit perturbation experiments. The generator set Eq. (4) consistently provided the most favourable diffusion behaviour among the candidates considered.

3.6. Primary State Update

The primary state evolves through a nonlinear update defined as

$$P_i^{(t+1)} = \text{RULEA}(P_i^{(t)}, P_{i-1}^{(t)}, P_{i+1}^{(t)}, P_{i+8}^{(t)}) \oplus P_{i+2}^{(t)}, \quad i \in \mathbb{Z}_{64}. \quad (5)$$

The additional term $P_{i+2}^{(t)}$ introduces a secondary coupling path that is independent of the four inputs to RULEA. During architecture validation, it was observed that the baseline design without this term occasionally exhibited cancellation behaviour in which state differences collapsed prematurely. The inclusion of the $+2$ offset reduced this effect while introducing only a single additional XOR operation.

Among candidate offsets $\{+1, +2, +4, +8, +16, +32\}$, the offsets $+2, +4, +16$, and $+32$ produced acceptable avalanche behaviour. Offset $+2$ was selected as the shortest viable offset and therefore the most favourable from a routing and hardware-efficiency perspective.

The update in Eq. (5) is applied simultaneously to all 64 positions using the current-round state bits, yielding a fully parallel combinational structure.

3.7. Auxiliary State Evolution and Injection

The auxiliary state evolves as a 64-bit Galois LFSR with feedback

$$fb^{(t)} = L_0^{(t)} \oplus L_1^{(t)} \oplus L_3^{(t)} \oplus L_4^{(t)}. \quad (6)$$

The register update is

$$L^{(t+1)} = (L_1^{(t)}, L_2^{(t)}, \dots, L_{63}^{(t)}, fb^{(t)}). \quad (7)$$

From the updated auxiliary state, an 8-bit round-dependent perturbation vector is extracted using a strided sampling pattern

$$rk^{(t+1)} = (L_0^{(t+1)}, L_8^{(t+1)}, L_{16}^{(t+1)}, L_{24}^{(t+1)}, L_{32}^{(t+1)}, L_{40}^{(t+1)}, L_{48}^{(t+1)}, L_{56}^{(t+1)}). \quad (8)$$

These bits are injected into the first eight positions of the primary state

$$P_j^{(t+1)} \leftarrow P_j^{(t+1)} \oplus rk_j^{(t+1)}, \quad j = 0, \dots, 7. \quad (9)$$

The strided extraction pattern ensures that the sampled bits are spread across the auxiliary register. The injection is intentionally simple and requires only fixed wiring in hardware.

3.8. LFSR Characteristic Polynomial and Period Analysis

The auxiliary state L evolves according to a linear feedback shift register (LFSR) defined over $\mathbb{F}_2$. The update rule is given by

$$L^{(t+1)} = (L_1^{(t)}, L_2^{(t)}, \dots, L_{63}^{(t)}, L_0^{(t)} \oplus L_1^{(t)} \oplus L_3^{(t)} \oplus L_4^{(t)}). \quad (10)$$

This corresponds to the linear recurrence

$$s_{t+64} = s_{t+4} \oplus s_{t+3} \oplus s_{t+1} \oplus s_t, \quad (11)$$

whose characteristic polynomial over $\mathbb{F}_2$ is

$$p(x) = x^{64} + x^4 + x^3 + x + 1. \quad (12)$$

The polynomial $p(x)$ was verified to be both irreducible and primitive over $\mathbb{F}_2$. In particular, irreducibility was established using a standard Frobenius-based test, and primitivity was confirmed by verifying that for each prime divisor q of $2^{64} - 1$,

$$x^{(2^{64}-1)/q} \not\equiv 1 \pmod{p(x)}. \quad (13)$$

The complete set of prime factors of $2^{64} - 1$,

$$2^{64} - 1 = 3 \cdot 5 \cdot 17 \cdot 257 \cdot 641 \cdot 65537 \cdot 6700417,$$

was used in this verification.

As a consequence, the LFSR defined by Eq. (10) has maximal period $2^{64} - 1$ for any nonzero initial state. This guarantees that the auxiliary state does not introduce short cycles or structural periodicity that could degrade the statistical properties of the keystream. The primitive LFSR therefore provides a sound linear backbone for the overall state evolution, while nonlinearity and security are introduced through the interaction with the primary state update.

3.9. Keystream Output Function

After the state update and injection stages, the design produces one keystream bit per round.

First, a nonlinear intermediate value is computed

$$r^{(t+1)} = \text{RULEA}(P_0^{(t+1)}, P_{16}^{(t+1)}, P_{32}^{(t+1)}, P_{48}^{(t+1)}). \quad (14)$$

The keystream bit is then defined as

$$z^{(t+1)} = r^{(t+1)} \oplus \bigoplus_{\substack{0 \leq i < 64 \\ i \notin \{0, 16, 32, 48\}}} P_i^{(t+1)}. \quad (15)$$

This structure balances nonlinearity and hardware cost. The nonlinear component provides degree-3 mixing at the output boundary, while the remaining state bits are aggregated through a lightweight XOR tree.

3.10. Round Structure

One round of SIR proceeds as follows:

1. Compute the tentative primary-state update using Eq. (5).
2. Update the auxiliary state using Eq. (6)–(10).
3. Extract the perturbation vector using Eq. (8).
4. Inject the perturbation bits into P_0 – P_7 using Eq. (9).
5. Generate the keystream bit using Eq. (14)–(15).

Steps 1 and 2 are independent and can be executed in parallel. The only data dependency occurs during the injection stage.

3.11. Initialization Phase

Before keystream generation, the internal state is initialized from the secret key and initialization vector (IV). Let the 128-bit key be $K = K^{(0)} \parallel K^{(1)}$, where $K^{(0)}$ and $K^{(1)}$ are 64-bit halves. Let V denote the 64-bit initialization vector. The primary and auxiliary states are initialized as

$$P^{(0)} = K^{(0)} \oplus V \quad (16)$$

$$L^{(0)} = K^{(1)} \oplus \text{RotL}_1(V) \oplus C \quad (17)$$

where $\text{RotL}_1(V)$ denotes a one-bit left rotation of V , and C is a fixed 64-bit constant defined as

$$C = 0 \times 9\text{E}3779\text{B}97\text{F}4\text{A}7\text{C}15. \quad (18)$$

The bit ordering follows the convention used throughout the paper, where the most significant bit corresponds to index 0 and the least significant bit corresponds to index 63.

After loading the initial state $(P^{(0)}, L^{(0)})$, the design performs a warm-up phase of 32 rounds in which the internal state evolves according to the round update rules described in Section 3.6, but no keystream bits are produced. This warm-up stage is intended to reduce direct dependence of the early keystream on the loaded key and IV and to allow diffusion to develop across the internal state before keystream generation begins.

For clarity and reproducibility, the complete initialization and keystream-generation procedure is summarized in Algorithm 1.

Algorithm 1 SIR initialization and keystream generation

Require: 128-bit key $K = K^{(0)} \parallel K^{(1)}$, 64-bit IV V , keystream length N

Ensure: Keystream sequence $z_0, z_1, \dots, z_{N-1}$

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1:  $P \leftarrow K^{(0)} \oplus V$ 
2:  $L \leftarrow K^{(1)} \oplus \text{RotL}_1(V) \oplus C$ 
   where  $C = 0 \times 9\text{E}3779\text{B}97\text{F}4\text{A}7\text{C}15$ 
3: for  $t = 0$  to 31 do
4:    $P' \leftarrow \text{UpdatePrimary}(P)$ 
5:    $L' \leftarrow \text{UpdateAuxiliary}(L)$ 
6:    $k \leftarrow \text{ExtractKey}(L')$ 
7:    $P \leftarrow \text{Inject}(P', k)$ 
8:    $L \leftarrow L'$ 
9: end for
10: for  $j = 0$  to  $N - 1$  do
11:    $P' \leftarrow \text{UpdatePrimary}(P)$ 
12:    $L' \leftarrow \text{UpdateAuxiliary}(L)$ 
13:    $k \leftarrow \text{ExtractKey}(L')$ 
14:    $P \leftarrow \text{Inject}(P', k)$ 
15:    $L \leftarrow L'$ 
16:    $z_j \leftarrow \text{Output}(P)$ 
17: end for
18: return  $z_0, z_1, \dots, z_{N-1}$ 

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In Algorithm 1, $\text{UpdatePrimary}(\cdot)$ denotes the 64-bit nonlinear parallel update of the primary state according to

$$P'_i = \text{RuleA}(P_i, P_{i-1}, P_{i+1}, P_{i+8}) \oplus P_{i+2}, \quad i = 0, 1, \dots, 63, \quad (19)$$

with all indices taken modulo 64. The auxiliary update $\text{UpdateAuxiliary}(\cdot)$ denotes the 64-bit LFSR transition

$$L' = (L_1, L_2, \dots, L_{63}, L_0 \oplus L_1 \oplus L_3 \oplus L_4). \quad (20)$$

The function $\text{ExtractKey}(L')$ returns the 8-bit vector

$$k = (L'_0, L'_8, L'_{16}, L'_{24}, L'_{32}, L'_{40}, L'_{48}, L'_{56}), \quad (21)$$

and $\text{Inject}(P', k)$ applies contiguous key injection into the first eight positions of the primary state:

$$P_i \leftarrow \begin{cases} P'_i \oplus k_i, & 0 \leq i \leq 7, \\ P'_i, & 8 \leq i \leq 63. \end{cases} \quad (22)$$

Finally, $\text{Output}(P)$ is defined as

$$z = \text{RuleA}(P_0, P_{16}, P_{32}, P_{48}) \oplus \bigoplus_{\substack{i=0 \\ i \notin \{0,16,32,48\}}}^{63} P_i. \quad (23)$$

The warm-up phase consists of 32 state-update rounds without keystream extraction, after which one keystream bit is produced per round.

The local interaction structure of the primary-state update is shown in Fig. 1. Each state bit is updated using the four-input nonlinear function RULEA applied to a neighbourhood defined by offsets $\{-1, +1, +8\}$ around the current position. An additional coupling term P_{i+2} is XORed with the nonlinear output to provide an extra cross-position dependency. All indices are interpreted modulo 64, and the update is applied in parallel to every bit position of the primary state.

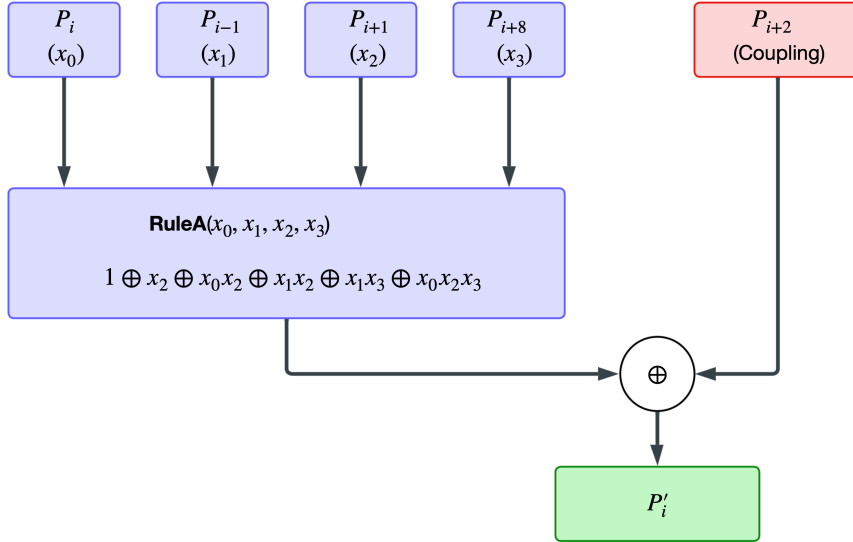


Figure 1: Local update rule for the primary state. For each position i , the nonlinear function RULEA is applied to the neighbourhood $(P_i, P_{i-1}, P_{i+1}, P_{i+8})$, and the result is XORed with the coupling term P_{i+2} to produce the tentative updated bit P'_i .

To clarify the interaction between the primary nonlinear state and the auxiliary LFSR, the overall architecture of SIR is illustrated in Fig. 2. The design consists of two interacting state components: a 64-bit nonlinear primary register P responsible for diffusion and nonlinear mixing, and a 64-bit auxiliary LFSR L that generates a small round-dependent injection vector. At each round, the primary state is updated through the sparse neighbourhood function RULEA, the auxiliary LFSR evolves linearly, and eight bits extracted from L are injected into the first eight positions of the primary state. The keystream bit is then derived from the updated primary state through a nonlinear output function.

3.12. Design Parameters

The principal architectural parameters of SIR are summarised in Table 1. The design maintains two 64-bit internal states, a primary nonlinear state P and an auxiliary linear state L , forming a combined 128-bit internal state. The design accepts a 128-bit secret key and a 64-bit initialization vector (IV). After initialization, the design performs a 32-round warm-up phase in which the internal state is iteratively updated without producing keystream output. Keystream generation then proceeds through repeated application of the nonlinear update rule, auxiliary-state evolution, and the output extraction function described in the preceding sections. Table 1 summarises these parameters and their roles in the overall architecture.

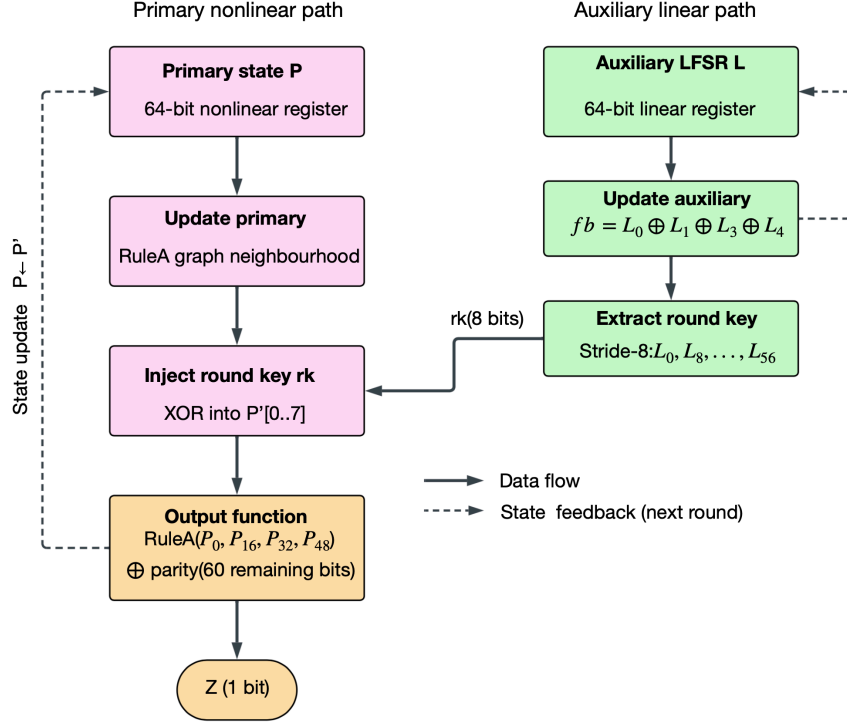


Figure 2: Overall architecture of SIR. The design maintains two interacting 64-bit registers: a nonlinear primary state P and a linear auxiliary LFSR L . Each round applies the nonlinear primary-state update, auxiliary LFSR transition, extraction of an 8-bit injection vector, contiguous injection into P , and nonlinear output extraction.

4. Diffusion and Statistical Analysis

This section presents an empirical evaluation of several structural properties of SIR. The analysis focuses on characteristics that are relevant to the practical behaviour of an iterative keystream generator, including the local properties of the nonlinear update rule, differential diffusion behaviour of the state, reduced-round differential screening, reduced-round linear-correlation screening, and state–output mutual-information analysis. These experiments are intended to examine how effectively local nonlinear interactions propagate through the sparse state-update structure and whether any observable statistical dependencies arise between the internal state and the generated keystream. All experiments were conducted using independent Python analysis scripts to ensure reproducibility.

The experiments reported here provide a systematic empirical structural evaluation of the construction, focusing on diffusion behaviour, statistical properties, and observable state–output dependencies. Their purpose is to identify potential structural weaknesses, such as insufficient diffusion, low-round differential or linear artefacts, or detectable statistical dependencies between state variables and keystream bits, that could undermine the credibility of the proposed construction. The absence of such weaknesses in the tested regime provides supporting empirical evidence for the structural robustness of the design.

Table 1: SIR design parameters.

Parameter	Value
Primary state size	64 bits
Auxiliary state size	64 bits
Total internal state	128 bits
Neighbourhood offset set	$\{-1, +1, +8\} \pmod{64}$
Local nonlinear rule	RULEA (degree 3, 4 inputs)
Secondary coupling offset	+2
LFSR feedback taps	$L_0 \oplus L_1 \oplus L_3 \oplus L_4$
Round-key extraction	$L_0, L_8, L_{16}, L_{24}, L_{32}, L_{40}, L_{48}, L_{56}$
Injection positions	$P_0, \dots, P_7$ (contiguous)
Output function	RULEA on $\{P_0, P_{16}, P_{32}, P_{48}\} \oplus$ parity of remaining 60 bits
Warm-up rounds	32
Keystream rate	1 bit per round

4.1. Properties of the Nonlinear Update Rule

The behaviour of each round depends in part on the local nonlinear mixing function RULEA defined in Eq. (3). Before examining the system-level behaviour, it is useful to record the intrinsic properties of this function, which can be computed exhaustively because it has only four input variables.

Table 2: Cryptographic properties of the RULEA Boolean function ($f : \mathbb{F}_2^4 \rightarrow \mathbb{F}_2$).

Property	Value
Number of input variables	4
Algebraic degree	3
Nonlinearity	4
Maximum Walsh coefficient	8
Maximum differential probability	0.75
Maximum linear bias	0.25

Table 2 summarises these properties. The function achieves algebraic degree 3, which is the maximum attainable for a four-variable Boolean function containing cubic terms. Its nonlinearity of 4 (equivalently, maximum Walsh coefficient of 8) and maximum differential probability of 0.75 are typical for balanced four-input Boolean functions and therefore do not, in isolation, provide strong resistance against classical differential or linear attacks. Consequently, RULEA is not intended to serve as a standalone cryptographic primitive. Instead, it is employed as a compact nonlinear coupling mechanism within an iterated state-update structure, where the overall behaviour arises from the interaction between local nonlinearity, sparse state connectivity, and repeated round-wise composition. The effective strength of the construction must therefore be evaluated at the system level rather than from the local properties of the Boolean function alone.

4.2. Differential Diffusion and Avalanche Behaviour

4.2.1. Experimental Setup

Diffusion was evaluated by a differential propagation experiment. Two instances were initialized with identical internal states except for a single-bit perturbation injected into the primary state P ; the auxiliary state L was kept

identical in both instances. The experiment was repeated over 500 independent trials with randomly generated initial states.

At each round t , the Hamming weight of the primary-state difference $\Delta P^{(t)}$ and the probability that the output bit differs between the two instances, $\Pr[\Delta z^{(t)} = 1]$, were recorded. These measurements provide an indication of how rapidly a local state disturbance propagates through the update structure.

4.2.2. Diffusion Profile

Table 3 reports the average Hamming weight of the primary-state difference and the corresponding output-difference probability at selected rounds in the single-bit perturbation experiment. The complete round-by-round diffusion behaviour during the initialization phase is illustrated in Fig. 3.

Table 3: Average Hamming weight of the primary-state difference ΔP and output bit difference probability following a single-bit perturbation (500 trials).

Round	Avg. $ \Delta P $	$\Pr[\Delta z = 1]$
1	2.75	0.464
2	4.85	0.462
3	7.59	0.506
4	10.37	0.520
5	13.20	0.516
6	15.78	0.458
7	18.48	0.500
8	21.01	0.492
9	23.69	0.524
10	26.08	0.480
15	31.12	0.512
20	31.79	0.494

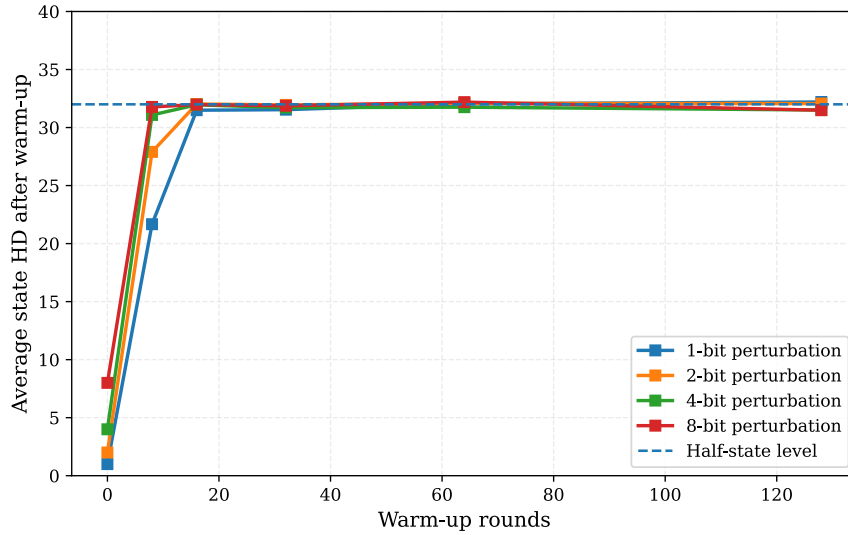


Figure 3: State mixing behaviour during initialization. A single-bit perturbation introduced in the primary state propagates across the internal state during the warm-up phase. The average Hamming distance approaches the half-state level, indicating rapid diffusion of differences.

The disturbance expands steadily across the primary state. The average difference grows from approximately three active bits after the first round to about twenty-six bits by round 10. By rounds 14–15 the average difference approaches 32 bits, which corresponds to the expected Hamming weight of a uniformly random difference over 64 bits. Beyond this point, the difference stabilizes near this level, indicating that the internal state has reached a broadly mixed regime. This behaviour is consistent with the sparse interaction pattern used in the state update, which allows local disturbances to spread across the state over repeated rounds.

The output difference probability $\Pr[\Delta z = 1]$ remains close to the ideal value of 0.5 throughout the experiment, indicating that even early-round state differences influence the keystream. No extinction events, rounds in which the state difference collapses to zero, were observed in any of the 500 trials. This behaviour contrasts with the earlier baseline architecture, where extinction occurred frequently. The revised update rule introduces additional cross-state interactions between neighbouring state positions, which reduces systematic cancellation of differences and promotes sustained propagation of disturbances through the state.

To further evaluate the avalanche behaviour of the construction, additional experiments were conducted using perturbations of different Hamming weights. The normalized keystream Hamming distance HD/n converges toward the ideal random value of 0.5 once a modest warm-up period is applied, as illustrated in Fig. 4. Across all tested perturbation weights, no experiment produced identical keystream sequences, indicating that extinction behaviour was not observed in the tested setting.

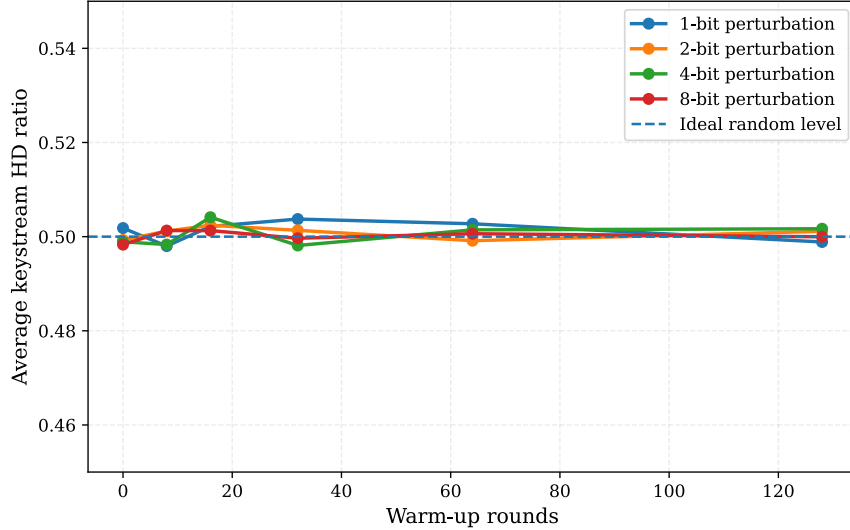


Figure 4: Avalanche convergence of the SIR architecture. Perturbations of different Hamming weights produce keystream differences approaching the ideal random level (0.5) after a short warm-up phase.

These observations support the use of a 32-round warm-up period from an empirical diffusion perspective. Since the diffusion horizon, the point at which the state difference resembles a random 64-bit vector, occurs around rounds 14–15, a warm-up of 32 rounds provides a practical margin before keystream generation begins. This reflects an empirical design choice guided by observed diffusion behaviour rather than a formal security bound.

4.3. Differential Propagation and Activity-Based Analysis

The local differential properties of RULEA (Table 2) indicate a maximum differential probability of 0.75, suggesting only moderate resistance when the function is considered in isolation. However, in the proposed design, the Boolean function is applied iteratively across a sparsely connected state network. To evaluate the resulting system-level behaviour, three complementary experiments were conducted focusing on differential propagation, extinction behaviour, and active nonlinear component growth.

4.3.1. Differential propagation and diffusion behaviour

A reduced-round differential propagation experiment was performed by injecting single-bit and low-weight differences into the primary state P , auxiliary state L , and their combinations. For each case, the average Hamming distance, difference coverage, and survival probability were evaluated over multiple independent trials.

Across all tested input differences, the probability that a nonzero difference persists remained close to unity throughout the evaluated rounds, with no observable systematic extinction. Moreover, the average primary-state Hamming distance increases rapidly, reaching approximately half of the state size (about 32 bits) within 12–16 rounds, while the difference coverage expands to all 64 state positions. These results indicate strong diffusion and confirm that local differences do not collapse or remain confined to small subsets of the state.

4.3.2. Absence of differential extinction

A key weakness in sparsely connected constructions is the potential for difference cancellation or extinction due to local interactions. To evaluate this, extinction probabilities were explicitly measured. For all primary-state input differences and most auxiliary-state cases, the probability of complete difference cancellation was negligible, with $p_{\text{nonzero}} \approx 1$ across all rounds. This demonstrates that the revised update structure successfully eliminates the cancellation behaviour observed in weaker variants and ensures persistent propagation of differences through the state.

4.3.3. Active nonlinear component growth and cumulative cost

To connect local properties of RULEA to global resistance, the number of active RULEA instances per round was measured under differential propagation. A local instance is considered active if at least one of its inputs differs between two evolving states.

For single-bit differences injected into the primary state, the number of active instances grows rapidly from approximately 4 at round 0 to about 24 by round 4 and exceeds 40 by round 8. In later rounds, nearly all 64 instances become active on average. Similar behaviour is observed for multi-bit differences, where activation occurs even more rapidly.

Using the exact local differential bound of RULEA, $\Pr_{\max} = 0.75$, an activity-based estimate of the cumulative differential cost can be obtained by associating a local weight $-\log_2(0.75)$ with each active RULEA instance. This leads to the expression

$$W_r^{(\text{est})} = \sum_{i=1}^r N_i \cdot (-\log_2(0.75)),$$

where N_i denotes the number of active instances in round i .

This formulation captures the contribution of nonlinear activity at the round level and provides a tractable abstraction of differential cost growth in the presence of structured state interactions. In the proposed architecture, neighbouring RULEA instances operate on partially overlapping input variables, and differential propagation therefore evolves through coupled interactions rather than independent components. The above expression reflects this behaviour at an aggregate level by linking cumulative cost directly to the expansion of active nonlinear elements.

Under this model, the cumulative cost grows rapidly with the number of rounds. For representative single-bit input differences, the estimated cumulative cost exceeds 90 by round 8 and continues to increase substantially in later rounds. This behaviour is consistent with the observed diffusion characteristics and highlights the rapid proliferation of active nonlinear interactions across the state.

4.3.4. Implications

These observations collectively indicate that, although the local Boolean function exhibits only moderate differential strength, the iterative architecture enforces rapid expansion of active nonlinear components and prevents difference extinction. As a result, the overall differential cost grows quickly with the number of rounds. This indicates that the structural behaviour of the design is governed by the interaction between sparse connectivity and iterative composition, rather than by the local properties of RULEA alone.

4.4. Reduced-Round Differential Screening

The previous experiment evaluates global diffusion behaviour. A complementary question is whether specific bit positions produce biased output differences at small round counts.

To examine this, a screening experiment was performed in which each of the 64 single-bit differences in the primary state was tested individually. For each difference position, the output difference probability was estimated over 2,000 trials for rounds 1 through 32.

Table 4 summarises the range of observed output difference probabilities across all tested positions.

Table 4: Reduced-round output differential screening results (2,000 trials per bit position).

Round	Max $\Pr[\Delta z = 1]$	Min $\Pr[\Delta z = 1]$	Max deviation
1	0.5310	0.4745	0.0310
5	0.5230	0.4720	0.0280
10	0.5215	0.4795	0.0215
15	0.5205	0.4825	0.0205
20	0.5310	0.4810	0.0310
25	0.5260	0.4625	0.0375
30	0.5185	0.4670	0.0330
32	0.5180	0.4715	0.0285

Across all tested rounds, the observed probabilities remain close to the ideal value of 0.5. The largest deviation observed in the experiment is approximately 0.038. Importantly, the bit position responsible for the maximum deviation varies from round to round, and no persistent differential pattern was observed within the tested set of single-bit input differences.

These observations suggest that the measured deviations are consistent with finite-sample fluctuation rather than a stable low-round differential artefact.

4.5. Reduced-Round Linear Correlation Screening

To complement the differential analysis, a linear screening experiment was conducted to detect possible correlations between the keystream output and simple linear functions of the initial state.

A structured family of candidate forms was evaluated, including individual bits of both registers, XOR pairs within each register, medium-range XOR combinations, and mixed primary–auxiliary forms. For each candidate function $\ell(P^{(0)}, L^{(0)})$, the empirical correlation with $z^{(r)}$ was measured over 20,000 random initial states for rounds 1 through 32.

Table 5 shows the maximum observed absolute correlation across all tested forms at selected rounds.

Table 5: Maximum observed absolute linear correlation (20,000 samples per round).

Round	Max $ \text{corr} $	Example form
1	0.0235	$P[9]$
5	0.0247	$L[15]$
10	0.0229	$L[3]$
15	0.0219	$P[15] \oplus L[15]$
20	0.0229	$P[21]$
27	0.0265	$L[2]$
32	0.0179	$L[61]$

The maximum observed correlation is 0.0265. For 20,000 samples, the expected standard deviation of a correlation estimate under independence is approximately $1/\sqrt{20000} \approx 0.007$. Consequently, values up to about 0.02 can arise purely from sampling fluctuation.

The candidate form achieving the largest correlation varies across rounds and shows no consistent structural pattern. This behaviour suggests that the observed values are dominated by statistical fluctuation rather than by a persistent simple linear relation between the tested state functions and the output. At the same time, this experiment provides a reduced-round screening of simple linear forms and complements broader correlation analysis.

4.6. State–Output Mutual Information Analysis

To obtain an information-theoretic view of state–output dependence, the mutual information between selected internal state bits and the output bit was measured. For each round r , the quantity

$$I(S_i^{(r)}; Z_r) = H(S_i^{(r)}) + H(Z_r) - H(S_i^{(r)}, Z_r) \quad (24)$$

was estimated using 20,000 random key–IV pairs.

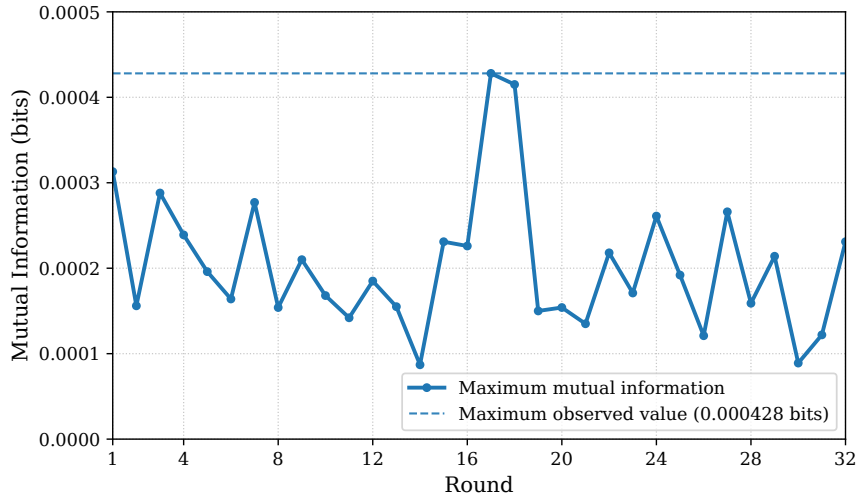


Figure 5: Maximum observed mutual information between tested internal state bits and the output bit.

Fig. 5 shows the maximum mutual information observed across all tested state positions at each round. The largest value recorded in the experiment is approximately 4.28×10^{-4} bits, with most rounds producing values around 10^{-4} bits.

These magnitudes are small and lie close to the estimation noise floor expected at this sample size. Furthermore, the state position producing the maximum value varies across rounds.

Within the resolution of this experiment, the results therefore do not reveal a stable information channel between individual internal state bits and the keystream output. As with the correlation analysis, this provides an empirical indication of limited first-order state–output dependence within the tested regime.

The experiments in this section collectively evaluate the diffusion and statistical behaviour of the proposed construction. Table 6 summarises the principal observations.

None of the experiments reveal obvious structural anomalies in the tested regime. The diffusion behaviour is consistent with the sparse interaction structure, while the correlation and information-theoretic analyses do not indicate strong dependence between the output and the tested simple subsets of the internal state. These observations provide useful empirical characterisation of the construction and complement broader cryptanalytic evaluation.

4.7. Round-Based Diffusion and Resistance Interpretation

The diffusion experiments presented in Section 4 characterize how local perturbations propagate through the internal state under iterative application of the update rule. While these results are empirical in nature, they provide useful insight into the structure of state evolution and its implications for resistance to statistical and structural attacks.

Table 6: Summary of empirical evaluation results.

Property	Observation
Diffusion horizon	State differences reach ≈ 32 active bits by rounds 14–15.
Differential screening	No persistent low-round output bias observed within the tested single-bit differences.
Linear correlation	Maximum observed $ \text{corr} = 0.0265$ with no stable dominant form among the tested linear expressions.
Mutual information	Maximum observed $I(S_i; Z_r) \approx 4.28 \times 10^{-4}$ bits.

Differential propagation experiments indicate that a single-bit perturbation in the primary state spreads to approximately half of the state variables within 14–15 rounds. This behaviour suggests that the influence of any individual state bit expands rapidly through the interaction network, resulting in a growing dependency set across iterations. In particular, after a moderate number of rounds, each state element is affected by multiple nonlinear combinations of prior state values, reducing the likelihood of low-weight differential characteristics persisting across many rounds.

From the perspective of statistical attacks, the observed rapid growth in active state variables contributes to the reduction of exploitable biases. Correlation and mutual-information measurements reported in Section 4 indicate negligible observable dependence between internal state components and the keystream output, with values close to the noise floor. This suggests that linear approximations and low-degree statistical relations do not remain stable under repeated iteration of the update function.

It is important to note that these observations do not constitute formal security bounds against specific attack classes such as differential or linear cryptanalysis. However, they provide empirical evidence that the sparse-interaction update mechanism promotes rapid mixing and expanding state dependencies, which are desirable properties for resisting distinguishing and correlation-based attacks.

Overall, the results indicate that the proposed architecture achieves effective diffusion through parallel combinational interaction, despite the absence of sequential shift-register propagation. This supports the view that sparse-interaction-based designs can provide sufficient mixing for practical security when combined with appropriate initialization and output extraction mechanisms.

4.8. Algebraic Degree Growth

Generators whose output function exhibits slow algebraic degree growth may be more exposed to algebraic modelling than constructions in which nonlinear dependencies accumulate quickly. Algebraic degree growth is therefore a useful structural indicator, although it is not sufficient in isolation to establish resistance against algebraic attacks.

To evaluate this property, the algebraic degree of the output bit $z^{(r)}$ was examined using a restricted input-subspace analysis. For a selected subset of input variables, the Boolean function mapping those variables to $z^{(r)}$ was evaluated exhaustively over the corresponding input space and converted to algebraic normal form (ANF). The algebraic degree of the resulting ANF was then recorded.

Experiments were conducted for input subspaces of dimension 6, 7, and 8 variables, and for up to 12 rounds of the update function. The observed degree growth is summarized below:

- the 6-variable subspace reaches degree 6 by round 3;
- the 7-variable subspace reaches degree 7 by round 4;
- the 8-variable subspace reaches degree 8 by round 8.

These results indicate that the algebraic complexity of the output function grows rapidly even in restricted projections of the state space.

In each case, the measured degree reaches the maximum possible for the corresponding subspace within a small number of rounds. Once the degree saturates, minor fluctuations may appear in subsequent rounds due to projection effects inherent to subspace-based measurements rather than genuine reductions in algebraic degree.

The rapid saturation of algebraic degree indicates that the nonlinear update rule, together with the chosen interaction pattern of the state update, quickly generates higher-order algebraic dependencies among the tested state variables. This behaviour is consistent with the diffusion properties observed in Section 4.2. However, because the analysis is based on restricted subspaces, it serves as a structural indicator of increasing algebraic complexity across rounds.

4.9. NIST SP 800-22 Statistical Tests

As a final empirical evaluation, the keystream generated by SIR was subjected to the NIST SP 800-22 statistical test suite [30]. A keystream of 10^8 bits was generated after a 32-round warm-up and partitioned into 100 non-overlapping sequences of 10^6 bits each. All fifteen major test categories of the SP 800-22 battery were applied.

For a sample size of 100 sequences and a significance level $\alpha = 0.01$, the minimum acceptable pass rate is 96 sequences for most tests and 55 out of 58 sequences for the Random Excursions variants. Table 7 summarises the results.

Table 7: NIST SP 800-22 statistical test results for SIR keystream (10^8 bits, 100 sequences of 10^6 bits).

Test	Proportion	P-value
Frequency (Monobit)	99/100	0.4944
Block Frequency	100/100	0.2897
Cumulative Sums (Fwd)	99/100	0.1626
Cumulative Sums (Rev)	99/100	0.7981
Runs	100/100	0.3041
Longest Run of Ones	99/100	0.8832
Binary Matrix Rank	99/100	0.4190
Discrete Fourier Transform	96/100	0.5955
Non-Overlapping Template	96–100/100	(all passing)
Overlapping Template	99/100	0.2023
Universal Statistical	99/100	0.2023
Approximate Entropy	100/100	0.7598
Random Excursions	$\geq 56/58$	(all passing)
Random Excursions Variant	$\geq 56/58$	(all passing)
Serial	100/100	0.9879
Linear Complexity	98/100	0.9879

All test categories satisfy the required pass-rate criteria. These results indicate that, at the level measured by the SP 800-22 battery, the generated keystream does not exhibit obvious statistical abnormalities. These tests therefore confirm the absence of observable statistical anomalies within the evaluated keystream sequences.

4.10. Long-Range Linear Complexity Profile

To obtain a more meaningful view of keystream linear complexity, the Berlekamp–Massey algorithm [31, 32] was applied to keystream prefixes of length up to 10^5 bits. The experiment was repeated over 20 independent random key–IV pairs, and the ensemble-average linear complexity was recorded at selected prefix lengths.

The measured profile remains extremely close to the random-sequence reference level $n/2$ throughout the tested range. For example, at $n = 10^5$ bits, the mean linear complexity is 50000.20, corresponding to a ratio of 0.500002 relative to sequence length, with a 95% confidence interval of [49999.93, 50000.47]. The width of this confidence interval is 0.54 bits, representing a deviation of less than 0.001% from the ideal reference level. This indicates that the ensemble variance across the 20 tested key–IV pairs is negligible relative to the sequence length, and that the linear complexity profile is highly stable across different initialisations.

Similar behaviour is observed across all intermediate prefix lengths, where the average linear complexity tracks the theoretical $n/2$ trend with only very small fluctuations consistent with finite-sample variation. The complete linear complexity profile across all evaluated prefix lengths is shown in Fig. 6.

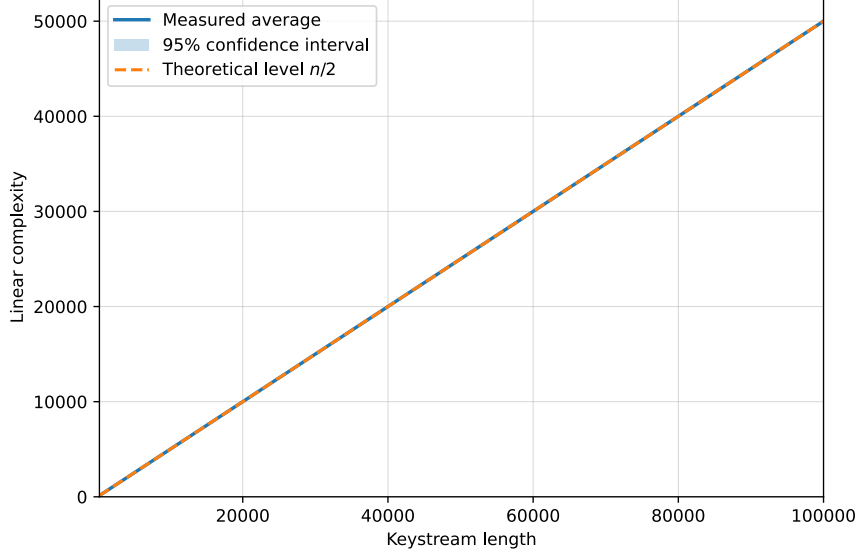


Figure 6: Ensemble-average linear complexity of generated keystream sequences measured over prefixes up to 10^5 bits (20 independent key-IV pairs). The shaded region denotes the 95% confidence interval of the ensemble estimate. The theoretical reference level $n/2$ for a uniformly random binary sequence is shown as a dashed line. The measured profile remains within the confidence band throughout the tested range, indicating no detectable short linear recurrence structure.

As shown in Fig. 6, the measured profile does not exhibit the systematic reduction in linear complexity relative to the $n/2$ reference that would indicate a short linear recurrence structure within the evaluated output length.

This is consistent with the high-degree algebraic mixing observed in Section 4.8 and with the absence of detectable simple linear correlations reported in Section 4.5.

The results should be interpreted as empirical evidence that the keystream does not exhibit detectable short-recurrence structure within the evaluated output length, rather than as a formal proof against all linear-complexity-based attacks. In particular, the analysis does not address linear complexity relative to the key or IV space, which would require a distinct algebraic treatment beyond the scope of the present empirical evaluation.

5. Hardware Evaluation

This section evaluates the hardware implementation of SIR and three established lightweight keystream generators, Grain-128 [6], Trivium [33, 34], and Espresso [8], under identical synthesis conditions. The evaluation proceeds in two parts. The primary analysis targets FPGA implementation on the Digilent Basys3 platform; a supplementary ASIC area analysis is then presented using an open-source standard-cell synthesis flow. Throughout, the same Verilog RTL used for FPGA synthesis was submitted directly to the ASIC tool, and no design-specific optimization for silicon implementation was performed.

5.1. FPGA Implementation Setup

All four designs were implemented on a Xilinx Artix-7 XC7A35TCPG236-1 device mounted on the Digilent Basys3 development board. The synthesis and implementation flow used Xilinx Vivado 2018.3 with default synthesis and implementation strategies. A single timing constraint of 10 ns (100 MHz clock) was applied uniformly to all designs. No design-specific placement constraints, floorplanning, or retiming directives were used. Resource utilization, timing slack, and on-chip power estimates were collected from the post-implementation reports generated by Vivado. Timing analysis is therefore post-route, reflecting placed-and-routed performance rather than pre-synthesis estimates.

The available device resources on the XC7A35T are 20,800 slice LUTs, 8,150 slices, and 41,600 flip-flops, providing a well-characterized reference for utilization percentages.

5.2. FPGA Resource Utilization

Table 8 summarizes the post-implementation resource utilization for all four designs. Absolute counts and percentages of the total available device resources are both reported.

Table 8: FPGA resource utilization on Xilinx Artix-7 XC7A35T (Basys3), Vivado 2018.3. Percentages in parentheses are fractions of total available device resources (LUTs: 20,800; Slices: 8,150; FFs: 41,600).

Cipher	Slice LUTs	Slices	Flip-Flops	Total Power (W)
SIR	183 (0.88%)	57 (0.70%)	177 (0.43%)	0.073
Grain-128	277 (1.33%)	81 (0.99%)	273 (0.66%)	0.083
Trivium	166 (0.80%)	46 (0.56%)	304 (0.73%)	0.069
Espresso	147 (0.71%)	45 (0.55%)	266 (0.64%)	0.075

SIR requires 183 slice LUTs and 57 slices, representing reductions of approximately 34% and 30%, respectively, compared with Grain-128. Its flip-flop count of 177 is similarly 35% lower than that of Grain-128. Against Trivium and Espresso, the picture is different: both of those designs rely on large NLFSR or register-shift structures, which increase their flip-flop counts (304 and 266, respectively) while keeping their LUT requirements lower. SIR occupies more LUTs than Trivium and Espresso, a consequence of the explicit combinational implementation of the RULEA update across 64 state positions, but uses significantly fewer flip-flops than either. This trade-off reflects the architectural difference between the sparse-interaction update used in SIR and conventional shift-register-based designs: a larger portion of the per-round mixing is implemented combinatorially in SIR rather than through sequential state shifting.

All four designs consume only a very small fraction of the available device resources, which indicates that the comparisons are not strongly influenced by placement pressure on the target device.

5.3. Timing Performance and Estimated Maximum Frequency

Table 9 reports the post-route timing results for all four implementations. The worst negative slack (WNS), worst hold slack (WHS), and worst pulse-width slack (WPWS) are taken directly from the Vivado timing reports. The estimated maximum clock frequency $f_{\max}$ is derived as

$$f_{\max} = \frac{1}{T_{\text{clk}} - \text{WNS}}, \quad (25)$$

where $T_{\text{clk}} = 10$ ns is the applied constraint period.

Table 9: Post-implementation timing results and estimated maximum clock frequencies (Artix-7 XC7A35T, 100 MHz constraint, Vivado 2018.3).

Cipher	WNS (ns)	WHS (ns)	WPWS (ns)	Est. $f_{\max}$ (MHz)	Throughput (Mbit/s)
SIR	4.539	0.141	4.500	183.1	183.1
Grain-128	4.762	0.116	4.500	190.9	190.9
Trivium	6.667	0.324	4.500	300.0	300.0
Espresso	6.667	0.324	4.500	300.0	300.0

All four designs meet the 100 MHz timing constraint with positive slack, confirming timing closure under the applied conditions. All WHS and WPWS values are positive, indicating that hold and minimum pulse-width constraints are satisfied without additional timing fixes.

SIR achieves an estimated $f_{\max}$ of approximately 183 MHz. This is slightly lower than Grain-128 (191 MHz) because the RULEA update involves a five-input combinational function (four neighbourhood bits plus the P_{i+2} cross-coupling term) that is replicated 64 times in parallel, producing a somewhat deeper critical path than Grain-128’s simpler tap-XOR feedback. Trivium and Espresso show substantially larger slack values, consistent with their shallower combinational paths, both rely primarily on shift-register feedback with a small number of tap-XOR operations, and consequently achieve an estimated $f_{\max}$ of 300 MHz under the same constraint. Since all four designs are iterative and produce one keystream bit per clock cycle, the throughput in Mbit/s equals $f_{\max}$ numerically.

It should be noted that the $f_{\max}$ values in Table 9 are estimates derived from post-route slack under a 100 MHz constraint, not the result of a dedicated maximum-frequency synthesis sweep. They nonetheless provide a consistent basis for comparison because identical tool settings were used for all designs.

5.4. Throughput, Latency, and Power

5.4.1. Throughput and Latency

Because all four designs follow an iterative, one-bit-per-cycle architecture, the steady-state keystream throughput equals the operating clock frequency. At the estimated maximum frequency of 183.1 MHz, SIR delivers a keystream rate of approximately 183 Mbit/s. Since the SIR wrapper assembles bits into bytes, the effective byte-rate throughput is

$$T_{\text{byte}} = \frac{f_{\max}}{8} \approx 22.9 \text{ MB/s.} \quad (26)$$

The initialization sequence consists of 32 warm-up rounds during which no output is produced, followed by 8 rounds to assemble the first complete keystream byte. Latency to the first ciphertext byte is therefore

$$L_{\text{first}} = \frac{40}{f_{\max}} \approx 218 \text{ ns at } 183 \text{ MHz.} \quad (27)$$

This initialization overhead is modest relative to the throughput and is consistent with the 32-round warm-up used in the design.

5.4.2. Power Consumption

Table 10 disaggregates the total on-chip power into dynamic and static components as reported by Vivado.

Table 10: On-chip power estimates from Vivado post-implementation reports (Artix-7 XC7A35T, 100 MHz operation).

Cipher	Total (W)	Dynamic (W)	Static (W)
SIR	0.073	0.002	0.072
Grain-128	0.083	0.002	0.072
Trivium	0.069	0.001	0.072
Espresso	0.075	0.003	0.072

The dominant component in every case is the device static power of 0.072 W, which is a fixed characteristic of the Artix-7 XC7A35T die at the nominal process corner and is largely independent of the implemented circuit. Dynamic power, the component attributable to switching activity within the core, is at most 3 mW across all four designs. SIR and Grain-128 share the same dynamic power estimate of 2 mW; Trivium is slightly lower at 1 mW, while Espresso is marginally higher at 3 mW. Because the dynamic component is very small relative to the static component, these FPGA power estimates should be considered in the context of the dominant static power component. In particular, power is not a strong differentiating metric at this scale; more meaningful dynamic-power comparisons would require either gate-level switching analysis or a silicon-oriented implementation flow.

Table 11: Consolidated FPGA comparison on Artix-7 XC7A35T. Throughput/LUT is computed as estimated $f_{\max}$ (Mbit/s) divided by slice LUT count.

Cipher	LUTs	FFs	Slices	$f_{\max}$ (MHz)	Throughput/LUT
SIR	183	177	57	183.1	1.00
Grain-128	277	273	81	190.9	0.69
Trivium	166	304	46	300.0	1.81
Espresso	147	266	45	300.0	2.04

5.5. Summary FPGA Comparison

Table 11 consolidates the key FPGA metrics and adds a throughput-per-LUT figure of merit that captures how efficiently each design converts combinational logic resources into keystream bits.

SIR achieves a throughput-per-LUT of 1.00 Mbit/s per LUT, which is substantially higher than Grain-128 (0.69) under the same flow. Trivium and Espresso show higher throughput-per-LUT figures owing to their shallower combinational paths and consequently higher estimated $f_{\max}$. At the same time, they rely on larger register banks (304 and 266 flip-flops, respectively), whereas SIR keeps the flip-flop count lower (177) at the cost of more combinational logic per cycle. The main observation is therefore an architectural trade-off rather than outright dominance: SIR occupies a point in the area-throughput design space characterised by moderate LUT usage together with a relatively small flip-flop count.

5.6. ASIC Synthesis Analysis

To obtain a technology-independent assessment of hardware cost, all four implementations were synthesized using the open-source Yosys synthesis framework (version used for all experiments) targeting the Nangate 45 nm Open Cell Library. The same Verilog source files used for FPGA implementation were submitted to Yosys without modification or ASIC-specific tuning, in order to preserve a direct correspondence between the two evaluations. This choice means that the ASIC figures provide an indication of architectural hardware cost without full technology-specific optimization; a production ASIC design would typically involve retiming, buffer insertion, and library-specific optimization not applied here.

Gate equivalents (GE) were computed by normalizing total cell area against the area of a single NAND2_X1 cell in the Nangate library ($A_{\text{NAND2}} = 0.798 \mu\text{m}^2$). No post-synthesis static timing analysis (STA) was performed, so maximum operating frequency figures are not reported for the ASIC scenario.

5.6.1. ASIC Synthesis Results

Table 12 presents the complete cell-level synthesis results for all four designs.

Table 12: ASIC synthesis results using Yosys and the Nangate 45 nm Open Cell Library ($A_{\text{NAND2}} = 0.798 \mu\text{m}^2$). Same unoptimized RTL as used for FPGA synthesis.

Cipher	Cells	FFs	Total Area (μm^2)	GE	Seq. Area (%)
SIR	1671	159	2457.31	3079	29.3
Grain-128	1151	268	2576.74	3228	55.3
Trivium	1259	303	2689.26	3369	59.9
Espresso	884	266	2118.69	2655	66.8

5.6.2. Area and Cell-Count Analysis

SIR occupies a total cell area of approximately $2457 \mu\text{m}^2$ (3079 GE), placing it between Espresso (the smallest at 2655 GE) and Grain-128 (3228 GE) and Trivium (the largest at 3369 GE). The design uses 1671 standard cells in total, the highest count among the four, because the 64-parallel instantiation of the RULEA function generates a larger

combinational logic cluster than the simple XOR-tap networks of Grain-128 and Trivium. Despite this, the total area is smaller than both Grain-128 and Trivium, indicating that the cells selected in the SIR netlist are, on average, smaller than some of the cells dominating those reference implementations.

5.6.3. Sequential Area Fraction: A Structural Signature

One of the most revealing metrics is the fraction of total area occupied by sequential elements. As shown in the final column of Table 12, the reference ciphers allocate a large majority of their silicon area to flip-flops: Espresso 66.8%, Trivium 59.9%, and Grain-128 55.3%. These values reflect the general design pattern of NLFSR/LFSR-based stream ciphers, where a large portion of the hardware budget is associated with register storage.

SIR exhibits a different profile. Only 29.3% of its area is sequential ($718.99 \mu\text{m}^2$, corresponding to 159 flip-flops), while the remaining 70.7% is combinational logic implementing the sparse interaction network. This shift in area composition is a direct hardware signature of the design style: rather than relying primarily on long shift-register structures, SIR places a larger fraction of the mixing logic in the combinational update network. The 159 flip-flops required correspond closely to the 128-bit internal state plus control registers, with minimal additional overhead.

5.6.4. Combinational Depth and Hardware Scalability

A practical consequence of the higher combinational fraction in SIR is that any increase in the number of state bits n would translate into additional combinational cells as well as additional flip-flops. In the reference ciphers, adding state bits also increases the register count and hence the sequential area. In SIR, the neighbourhood remains bounded, so increasing n adds $O(n)$ RULEA instances and $O(n)$ flip-flops. This suggests that the overall structural balance between combinational and sequential logic may remain qualitatively similar for larger state sizes, although a full characterization of scaling behaviour is reserved for future work.

5.6.5. Caveats on the ASIC Analysis

Several important caveats apply to the ASIC results presented here. First, the Yosys synthesis was performed without post-synthesis STA, timing-closure-driven retiming, or library-specific cell selection; consequently, no ASIC maximum operating frequency is reported. Second, the Nangate 45 nm library is an open academic library and does not reflect the characteristics of any commercial process node; the absolute area figures should therefore not be compared directly with published ASIC results that used commercial PDKs. Third, the RTL was not adapted for silicon targets: constructs that synthesize efficiently in Vivado (e.g., wide XOR trees inferred from Verilog reduction operators) may not be optimal under ASIC synthesis. The ASIC section is therefore presented as a structural and relative area comparison, not as a silicon implementation claim.

The hardware evaluation results, taken together, reveal a consistent picture of SIR’s implementation characteristics.

On the FPGA, the most notable result is the reduction in LUT and flip-flop count relative to Grain-128 under identical synthesis conditions. The 34% reduction in LUT usage and 35% reduction in flip-flop count are meaningful for compact implementations. The throughput-per-LUT figure of 1.00 Mbit/s/LUT is 45% higher than Grain-128’s 0.69, indicating that the SIR update structure compares favourably with Grain-128 in this particular implementation setting.

The comparison with Trivium and Espresso highlights a different architectural trade-off. Both ciphers achieve lower LUT counts by relying on larger register arrays, producing higher flip-flop counts in exchange. Accordingly, SIR should not be viewed as uniformly superior to these references, but rather as occupying a different region of the implementation design space.

On the ASIC side, SIR’s area of 3079 GE is within the range of the compared lightweight ciphers (2655–3369 GE), indicating that the proposed architecture does not incur a large area penalty under the adopted flow. The low sequential fraction (29.3% versus 55–67% for the references) is the most distinctive ASIC finding: it shows that the design relies less heavily on sequential storage and more heavily on combinational update logic than the compared shift-register-based ciphers.

The iterative, one-round-per-clock architecture used here was selected primarily for implementation clarity, direct verification against the reference model, and low area. It is not the only viable mapping: a 2×- or 4×-unrolled variant would increase throughput at a corresponding increase in LUT cost, following a conventional area–throughput trade-off. Exploring this design space is a natural direction for future work.

6. Security Analysis

This section discusses the security considerations of SIR with respect to several attack classes relevant to iterative keystream generators. The discussion is intentionally scoped: SIR is a new construction, and the present work does not claim a complete formal proof of security against all adversarial strategies. The assessment below is grounded in the empirical evidence of Section 4, supplemented by structural observations about the design. Where direct analysis has not been performed, this is stated explicitly.

The presentation is organized by attack class in order to distinguish between aspects of the design that are supported by experimental evidence, aspects that are supported primarily by structural reasoning, and aspects that remain open for future investigation.

6.1. Security Goals and Claimed Level

SIR maintains a 128-bit internal state, divided equally between the 64-bit primary nonlinear state P and the 64-bit auxiliary LFSR L . The design uses a 128-bit key together with a 64-bit IV. Accordingly, the natural generic target is resistance to exhaustive key search at the 128-bit level. No claim beyond this generic target is made.

The primary security objective is keystream indistinguishability: the output sequence $\{z^{(t)}\}_{t \geq 0}$ should not exhibit practical structural weaknesses that permit efficient distinguishing or state recovery. The arguments in this section should therefore be understood as design-oriented security considerations supported by empirical observations, rather than as formal proofs.

6.2. Exhaustive Key Search

Under a fixed IV, exhaustive search over the 128-bit key requires 2^{128} candidate evaluations, which is infeasible under conventional computational assumptions. The initialization mapping combines the key and IV through XOR and rotation operations followed by 32 rounds of state evolution before keystream generation begins. No shortcut attack on the key-loading structure emerges from the present analysis.

The current evidence therefore does not indicate an evident weakness in the initialization mapping that would reduce brute-force complexity below the generic key-search bound.

6.3. Time–Memory–Data Tradeoff Attacks

Time–memory–data tradeoff (TMDTO) attacks exploit the internal state size to launch distinguishing or state-recovery attacks below the exhaustive-search bound. Classical Hellman-style tradeoffs and their extensions to keystream generators, such as the Biryukov–Shamir framework [35], become less attractive as the internal state increases and as the (K, IV) pair is not reused.

SIR has a 128-bit internal state, so the generic TMDTO barrier is set by that state size. As with other keystream generators, safe usage therefore requires fresh IV values across sessions. Under that standard operational assumption, the state size does not suggest an immediate TMDTO concern from a generic viewpoint.

One structural caution is worth noting. The auxiliary state L evolves linearly as a 64-bit LFSR and would be predictable in isolation. In the actual construction, however, it is not directly observable and is coupled to the nonlinear primary state through the injection mechanism. This coupling complicates a clean state-separation attack, although the present work does not provide a formal complexity analysis for such a strategy. The absence of an evident TMDTO weakness should therefore be understood as a structural observation rather than a definitive negative result.

6.4. Correlation Attacks

Correlation attacks on keystream generators, beginning with Siegenthaler’s work [36] and extended to fast correlation attacks by Meier and Staffelbach [37], exploit statistical dependence between the keystream output and some linear function of the internal state. Such attacks are especially effective when the generator decomposes into linearly evolving components connected by a weak combining function.

SIR does not match the classical architecture assumed by those attacks: it does not consist of independent LFSRs feeding a simple combining function. Nevertheless, the possibility of observable correlation with simple linear functions remains relevant. The screening experiment of Section 4.5 was intended as a first check of this possibility.

Across the tested set of simple linear forms, the observed correlations were small and did not exhibit a stable dominant pattern across rounds.

These observations are encouraging, but they should not be overstated. The analysis considered only a limited family of candidate linear expressions and a moderate sample size. It therefore supports the claim that no obvious low-order linear relation was detected in the tested regime, but it does not rule out more sophisticated correlation attacks.

The output function (15) introduces a nonlinear component through RULEA, so the keystream is not a purely linear function of the state. This provides a structural obstacle to classical correlation strategies, although the strength of that obstacle has not been quantified formally in the present work.

6.5. Differential Cryptanalysis and Distinguishing Attacks

Differential analysis of keystream generators may target either the initialization process or the running generator. In both settings, the core question is whether particular input differences propagate to the output with exploitable bias.

6.5.1. Internal State Differential Behavior

The experiments of Section 4.2 examine this question for primary-state perturbations. In the tested setting, single-bit differences introduced into P expand rapidly across the state, and the observed output-difference probability remains close to $1/2$. The reduced-round screening of Section 4.4 similarly did not reveal a persistent output bias among the tested single-bit input differences over rounds 1–32.

These results suggest that no obvious low-round differential artefact is present in the examined regime. However, the experiments do not amount to an exhaustive search for differential characteristics, nor do they address all possible combined differences across both state components.

6.5.2. On the $\Delta L^{(0)} = 0$ Evaluation Setting

Most of the reported diffusion experiments were performed with $\Delta L^{(0)} = 0$ and a nonzero primary-state difference. This choice is useful for isolating the behaviour of the primary update, but it does not cover the full mixed differential space.

If $\Delta L^{(0)} \neq 0$, then the injection mechanism introduces additional round-dependent perturbations into the first eight positions of the primary state. Structurally, these perturbations are limited in width at each round, and once they activate differences in P , the nonlinear update tends to spread them. This suggests that the primary update should dominate the subsequent difference evolution. Even so, a dedicated mixed-state differential study has not been performed here.

Accordingly, the current evidence supports the absence of obvious differential weakness in the tested single-bit primary-state setting, while leaving the more general mixed differential case as an open problem.

6.6. Linear Cryptanalysis

Linear cryptanalysis seeks linear approximations of the form $\ell(P^{(0)}, L^{(0)}) \approx z^{(r)}$ with non-negligible bias. The reduced-round screening experiment of Section 4.5 evaluated a structured family of simple linear forms over 20,000 random initial states. The maximum observed correlation across the tested set and rounds 1–32 was small and did not remain associated with a stable form across rounds.

This is consistent with the absence of an obvious simple linear approximation in the tested family. It does not, however, amount to a complete linear cryptanalysis of the design. The search space of possible masks is far larger than the subset evaluated here, and the sample size is sufficient only for detecting relatively coarse bias.

The output function contributes some structural resistance by including a nonlinear RULEA term rather than a purely linear parity expression. This means that a trivial single-mask approximation is not immediately exposed by the output layer alone. Any stronger claim about resistance to linear cryptanalysis would require more systematic search methods than those used in the present work.

6.7. Algebraic Attacks

Algebraic attacks on keystream generators, introduced by Courtois and Meier [38] and extended to fast algebraic variants [39], attempt to represent the keystream as a system of low-degree multivariate equations in the initial state or key variables. Their practicality depends heavily on how slowly algebraic degree grows through initialization and state evolution.

The empirical algebraic-degree study reported in Section 4.8 and illustrated in Fig. 7 examined the degree of $z^{(r)}$ over restricted input subspaces.

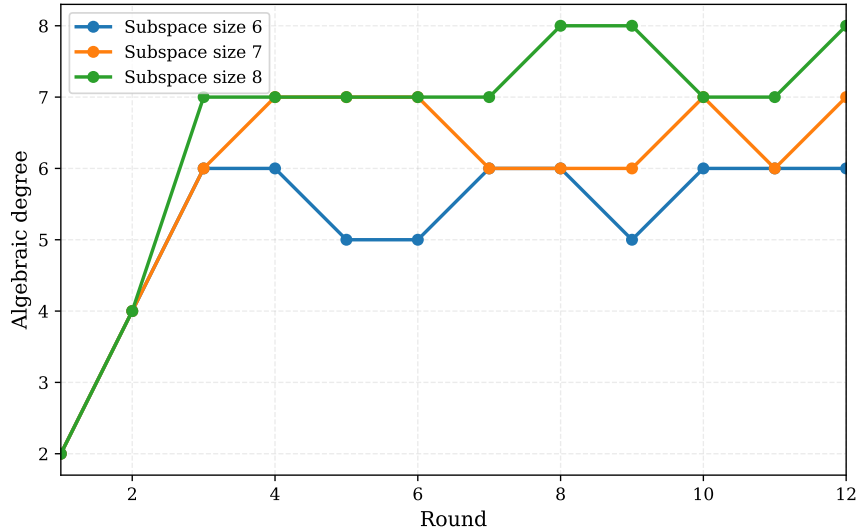


Figure 7: Algebraic degree growth of an output bit across rounds. The degree increases rapidly with round count, indicating fast growth of nonlinear dependencies within the tested subspaces.

6.7.1. Degree Growth of the Output Function

Within the tested subspaces of dimensions 6, 7, and 8, the ANF degree of $z^{(r)}$ reached the subspace maximum after a small number of rounds. This indicates rapid growth of algebraic complexity in those restricted projections.

That observation is useful, but it must be interpreted carefully. Subspace saturation does not prove that the full 64-variable function is free of exploitable low-degree structure. It only indicates that within the examined projections, the output function does not remain shallow for many rounds. Accordingly, the experiment provides supportive structural evidence against very low-degree behaviour, but not a definitive negative result for algebraic attack.

6.7.2. Cube Attacks

Cube attacks [40] are especially relevant to keystream generators with key/IV initialization, since they exploit low-degree polynomial structure in the output as a function of IV variables and key bits.

The rapid degree growth observed in the restricted-subspace experiments suggests that the 32-round initialization is likely to generate substantial algebraic complexity before any output bit is produced. This is a favourable indicator, but it is not a substitute for a direct cube-attack analysis.

A dedicated cube evaluation, for example, enumerating IV cubes of various dimensions and measuring the resulting superpolies, has not been performed in this work. Cube attacks should therefore be regarded as an explicitly open topic rather than as a closed issue.

6.8. Guess-and-Determine Attacks

Guess-and-determine (GaD) attacks attempt to guess a subset of internal state variables and derive the remainder from the update and output equations. Their effectiveness depends on how much of the state must be guessed before the remaining variables become tractable.

In SIR, each primary-state update bit depends on multiple nearby state bits through a nonlinear degree-3 function together with an additional coupling term. This structure does not appear to support a simple forward-determination procedure from a small guessed subset. In addition, the auxiliary LFSR contributes round-dependent perturbations to the primary state, so the two components do not separate cleanly under partial guessing.

These features suggest that straightforward low-cost GaD strategies are unlikely. However, no formal GaD analysis or tight complexity bound has been derived. This attack class should therefore be considered only qualitatively discussed in the present manuscript.

6.9. Related-Key and Related-IV Security

Related-key attacks [41] and related-IV attacks examine whether structured relations in the initialization inputs lead to correlated keystream behaviour.

SIR initializes $P^{(0)} = K^{(0)} \oplus V$ and $L^{(0)} = K^{(1)} \oplus \text{RotL}_1(V) \oplus C$, so any IV difference induces structured initial differences in both state components. The warm-up process is intended to mix these differences before keystream generation begins.

The diffusion experiments suggest that state differences spread rapidly during the early rounds, which is favourable from the perspective of related-IV resistance. Even so, no dedicated related-IV or related-key attack analysis has been performed in this work. In particular, the structured form of the initialization mapping means that such analysis would be worthwhile in future work. This topic should therefore be regarded as open.

6.10. Distinguishing Attacks via Statistical Testing

A distinguishing attack attempts to separate the generator output from a uniformly random sequence using a feasible amount of keystream. Section 4.9 reported that SIR passes the NIST SP 800-22 statistical tests on 10^8 bits of output. In addition, the linear-complexity profile of the generated keystream was evaluated over sequences of up to 10^5 bits using the Berlekamp–Massey algorithm, with results reported in Section 4.10. The measured profile remains extremely close to the theoretical $n/2$ reference throughout the tested range, providing no indication of a short linear recurrence that would support a practical linear-complexity-based distinguisher within the evaluated output length.

Beyond the NIST battery, the mutual-information analysis of Section 4.6 provides a complementary first-order view of state–output dependence. The measured values were very small in the tested setting and did not reveal a stable leakage channel associated with individual state bits.

These observations are useful as statistical screening, but their scope should be stated precisely. Passing SP 800-22 and exhibiting small first-order mutual information do not prove cryptographic security, nor do they exclude higher-order distinguishers. They do indicate that the generator does not exhibit obvious large-scale statistical abnormalities in the tested regime.

Table 13 consolidates the present security discussion by attack class, indicating the type of evidence currently available.

Taken together, the available evidence suggests that SIR does not exhibit obvious low-order statistical, differential, or linear weaknesses within the experimental range examined in this work. However, several important topics remain open, including cube attacks, formal guess-and-determine analysis, and dedicated related-IV study. Accordingly, the present section should be read as a bounded and explicitly scoped security discussion rather than as a comprehensive cryptanalytic validation of the construction.

7. Discussion

The results presented in the previous sections provide an empirical characterization of the structural, statistical, and implementation properties of the proposed SIR architecture. In contrast to conventional lightweight keystream generators that rely on shift-register-based state propagation, the present design employs a sparse nonlinear state-interaction network coupled with an auxiliary linear component. This distinction enables a direct assessment of how replacing sequential propagation with parallel interaction influences diffusion behaviour, statistical characteristics, and hardware cost.

Table 13: Summary of security arguments by attack class. Evidence codes: E = experimental evidence from this work; S = structural argument; O = open/not analyzed.

Attack class	Key supporting observation	Evidence
Exhaustive key search	128-bit key; no evident shortcut from current initialization analysis.	S
TMDTO attacks	128-bit state; standard fresh-IV assumption.	S
Correlation attacks	No stable simple low-order correlation detected in the tested forms.	E+S
Linear cryptanalysis	Small measured correlations among the tested linear expressions; no stable dominant form.	E+S
Differential distinguishing	Rapid diffusion in tested primary-state experiments; no persistent low-round output bias observed in tested cases.	E+S
Algebraic attacks	Rapid degree growth in restricted subspaces.	E+S
Cube attacks	No direct cube analysis performed.	O
Guess-and-determine	Structurally nontrivial due to nonlinear coupling, but no formal complexity bound derived.	S / O
Related-key / related-IV	No dedicated analysis performed.	O
Distinguishing (statistical)	NIST SP 800-22 passed (10^8 bits); small first-order mutual information in tested experiments.	E

7.1. Diffusion Characteristics

The differential propagation experiments show that the design develops broad state diffusion within a modest number of rounds. Starting from a small perturbation, the average number of active state differences grows steadily and approaches approximately half of the primary-state width after about 14–15 rounds. The corresponding output-difference probability remains close to the ideal random reference level in the tested setting.

These observations indicate that the update mechanism enables rapid expansion of local disturbances through repeated nonlinear interaction. From a structural perspective, this behaviour suggests that dependency sets grow across rounds, reducing the persistence of low-weight differential characteristics. Although the experiments do not constitute a complete differential cryptanalysis, they provide evidence that the architecture achieves effective diffusion without relying on sequential shift-register propagation.

7.2. Linear Correlation Behaviour

The reduced-round correlation experiments did not reveal a stable simple linear relation between the tested state expressions and the keystream output. The measured correlations remained small and varied across rounds, which is consistent with the absence of a dominant low-order linear approximation in the examined setting.

While the evaluated mask set is limited and does not represent a complete linear cryptanalysis, the results indicate that simple linear structures do not persist under iteration of the update function. This behaviour is consistent with the increasing nonlinear dependency introduced by the interaction-based update mechanism.

7.3. State–Output Information Leakage

The mutual-information measurements provide a complementary information-theoretic perspective on first-order state–output dependence. The observed values were close to the noise floor in the tested regime, indicating that no direct single-bit statistical leakage was detected between individual state variables and the output.

These results do not exclude higher-order dependencies or more complex distinguishers. However, they provide initial evidence that the generator does not expose trivial first-order leakage channels, which are typically screened during early-stage evaluation of keystream constructions.

7.4. Hardware Implementation Characteristics

The hardware evaluation highlights a central property of the proposed architecture. On the FPGA platform, SIR uses fewer LUTs and flip-flops than Grain-128 under the adopted implementation flow, while remaining within the same general compact-hardware regime as the other reference designs. The comparison with Trivium and Espresso shows that the design is not uniformly optimal, but instead occupies a distinct point in the trade-off space between combinational logic and sequential storage.

This distinction becomes more apparent in the ASIC synthesis results. SIR exhibits a comparatively lower sequential-area fraction and a higher combinational share than the reference shift-register-based designs. This is a direct consequence of the interaction-based update mechanism, which shifts part of the diffusion process from sequential propagation to combinational logic.

From a hardware-design perspective, this behaviour is significant. It indicates that the proposed architecture enables a redistribution of cost between storage and logic resources, which may be advantageous in platforms such as FPGAs where logic and register utilization are not equally constrained.

7.5. Architectural Interpretation

A key implication of the present work is that diffusion in lightweight keystream generators need not be intrinsically tied to sequential propagation. In conventional shift-register-based designs, diffusion is achieved through the gradual movement of state information across the register, coupling the rate of mixing to the underlying dataflow.

In contrast, the SIR construction demonstrates that diffusion can be realized through parallel sparse interaction among state variables. Under iterative application, local nonlinear interactions generate expanding dependency structures, enabling global mixing without requiring explicit data movement across long registers.

From an architectural viewpoint, this represents a decoupling of diffusion from storage. The state no longer acts solely as a medium for propagation but as a substrate for interaction. This shift leads to a distinct hardware profile in which reduced sequential state is exchanged for increased combinational mixing.

Rather than replacing existing designs, this approach introduces an additional design dimension for lightweight keystream generators. It enables exploration of implementation regimes that are not naturally accessible within the shift-register paradigm, particularly in contexts where logic resources are relatively inexpensive compared to storage or routing complexity.

7.6. Limitations and Future Work

Several limitations should be stated explicitly. First, the security evaluation is primarily empirical and does not include dedicated cube-attack experiments, formal guess-and-determine complexity analysis, or related-IV cryptanalysis. These remain important directions for future work in establishing stronger security assurances.

Second, the ASIC evaluation is based on a standard open synthesis flow without timing closure or silicon-level optimization. The reported results should therefore be interpreted as structural comparisons rather than finalized implementation metrics.

Third, the study focuses on a single parameterization of the architecture. Variations in neighbourhood structure, auxiliary-state coupling, nonlinear update functions, and initialization procedures may lead to different security and hardware trade-offs, and warrant systematic investigation.

Overall, the results indicate that SIR is a compact hardware-oriented keystream generator whose sparse nonlinear state-interaction structure achieves effective empirical diffusion and competitive implementation cost. While comprehensive cryptanalytic validation remains an open task, the present study establishes the architectural viability of interaction-based keystream generation as a meaningful alternative to conventional shift-register-based designs.

8. Conclusion

This work investigated sparse-interaction state evolution as an alternative architectural basis for lightweight keystream generation. Whereas conventional lightweight stream generators typically realise diffusion through sequential shift-register propagation, the proposed approach explores a different design principle in which mixing is produced through repeated local nonlinear interaction among state variables. The resulting architecture decouples diffusion from register-based propagation and therefore leads to a different balance between combinational logic and sequential storage.

Based on this principle, SIR was introduced as a lightweight keystream generator composed of a 64-bit nonlinear primary state and a 64-bit auxiliary linear state. The primary state evolves through a compact local Boolean rule applied over a sparse neighbourhood, while the auxiliary LFSR provides round-dependent perturbation. This combination was designed to achieve effective iterative mixing while preserving a compact and hardware-oriented implementation structure.

The empirical evaluation indicates that the proposed construction achieves rapid state mixing and does not exhibit obvious low-order structural weaknesses within the scope of the performed experiments. Differential propagation shows that local perturbations spread to approximately half of the primary state within 14–15 rounds. Reduced-round correlation screening did not reveal stable simple linear relations among the tested expressions, and mutual-information analysis produced values close to the estimation noise floor, with a maximum below 4.28×10^{-4} bits. Together with the NIST SP 800-22 results and long-range linear-complexity measurements, these observations support the empirical soundness of the construction in the tested regime.

The hardware evaluation demonstrates the practical significance of this architectural choice. FPGA implementation on a Xilinx Artix-7 (Basys-3) platform requires 183 LUTs and 177 flip-flops, confirming a compact realization. Comparative evaluation against Grain-128, Trivium, and Espresso shows that SIR occupies a distinct implementation point characterised by reduced sequential state and increased combinational mixing. The ASIC synthesis results exhibit the same structural trend, particularly in the substantially lower sequential-area fraction relative to conventional shift-register-based designs.

The main contribution of this work is therefore not merely the introduction of another lightweight keystream generator, but the demonstration that sparse-interaction-based diffusion can form a viable hardware-oriented design strategy. The results show that effective mixing need not be tied to sequential propagation, and that separating diffusion from storage leads to a coherent and practically relevant implementation profile.

From a broader design perspective, the proposed architecture introduces an additional point in the lightweight keystream design space. It may be especially relevant in FPGA-oriented and embedded settings where the trade-off between flip-flop usage, routing complexity, and per-cycle combinational mixing differs from that of conventional stream-cipher architectures.

Several directions remain open for future work. A more extensive cryptanalytic investigation, including cube attacks, broader differential and linear analysis, and formal guess-and-determine complexity evaluation, would strengthen the security assessment. Further exploration of neighbourhood structures, nonlinear update functions, auxiliary-state coupling mechanisms, and state sizes may also reveal improved security–hardware trade-offs. Finally, silicon-oriented optimization, energy characterization, and side-channel resistance remain important topics for practical deployment.

Declaration of Competing Interests

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Codes Availability

The reference implementation and test vectors are publicly available at:

<https://github.com/susantha-wijesinghe/sir-keystream-generator>

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